

Chapter 1 - Just enough category theory to be dangerous

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1 Categories and functors

Exercise 1.A

1. Since there's only one object, namely e , in the category, all the morphisms in such category will send e to e .
 - The combinations of morphisms are always associative.
 - There must exist an identity morphism, which'll be the identity in the group.
 - Since it's a groupoid, every morphism has a inverse.

(In fact every group can be represented using the language of category. Consider Cayley's Theorem and we can see every group element can be regarded as an automorphism, satisfying the case of morphism to a object itself.)

2. We consider a category $\mathcal{C}$ whose objects are A, B . Consider

$$\text{Mor}(\mathcal{C}) = \{\text{Id}_A, \text{Id}_B\}.$$

and we can see that it's a groupoid where the elements corresponding to Id_A, Id_B cannot have multiplication, therefore not a group.

Exercise 1.B

1. Similar to the case in **1.1.A**, we can use exactly the same proof to show that invertible elements of $\text{Mor}(A, A)$ forms a group.
2. Automorphism groups of objects in **Example 1.1.2** are those permutation groups permutating the elements in a group.
3. Automorphism groups of objects in **Example 1.1.3** are those invertible linear transformations mapping a linear space to itself.

Exercise 1.C [Skipped, require 1.1.21 to solve.]

- (i) In the case of finite cases, the set of bases of $\text{Vec}_k, \text{Vec}_k^\vee$ is finite. Therefore the functor $(\cdot)^\vee$ must be isomorphism.

Exercise 1.D

Since it's finite, just find out the bases and we can define a trivial isomorphism between $\mathcal{V}$ and $f.d.\text{Vec}_k$ by mapping those bases respectively.

2 Universal properties determine an object up to unique isomorphism

Exercise 2.A

1. For any two initial objects A, B , we define

$$\varphi : A \mapsto B, \psi : B \mapsto A.$$

And we can see

$$\varphi \circ \psi : B \mapsto B, \psi \circ \varphi : A \mapsto A.$$

Also, notice since A, B are initial objects, they have only one unique map to every objects in the category, including themselves. And Id_A, Id_B maps A, B to themselves. Hence we obtain

$$\begin{aligned} Id_A &= \psi \circ \varphi \\ Id_B &= \varphi \circ \psi. \end{aligned}$$

So we know φ, ψ are isomorphisms. □

2. The proof is similar to (a), skipped.

Exercise 2.B

1. In the category of sets, the initial and final object is $\emptyset$ and the class of all sets, respectively.
2. In the category of rings, the initial and final object is the zero ring and the class of all rings, respectively.
3. *Skipped (haven't learn topology yet).*

Exercise 2.C

- ($\Leftarrow$): When S contains no zerodivisors, we prove by contradiction: Assume that under the canonical ring map $\varphi : A \mapsto S^{-1}A$, there exist $a_1, a_2 \in A$, s.t. $\varphi(a_1) = \varphi(a_2)$. Now we have $a_1/1 = a_2/1$, which means

$$\exists s \in S, \text{ s.t. } s(a_1 - a_2) = 0.$$

Which contradicts with the fact that S has no zerodivisors!

- ($\Rightarrow$): When $\varphi : A \mapsto S^{-1}A$ is injective, we assume S contains a zerodivisors s at which

$$\exists d \in A, \text{ s.t. } d \neq 0, s \cdot d = 0.$$

Now let $a_1, a_2 \in A$, s.t. $a_1 - a_2 = d$. Since φ is injective, we must have $\varphi(a_1) \neq \varphi(a_2)$. Which contradicts with the fact that

$$s(a_1 - a_2) = s \cdot d = 0.$$

□

Exercise 2.D

Suppose

$$\begin{aligned} f : A &\mapsto B \\ \varphi : A &\mapsto S^{-1}A. \end{aligned}$$

By the first translation in by author, we have to prove that there an unique ring homomorphism ψ mapping $S^{-1}A$ to B . Recall that $\forall a \in A, f(a) = \varphi(a/1)$. Also, for all $a \in A, s \in S$,

$$\frac{a}{s} \cdot \frac{s}{1} = \frac{a}{1}.$$

So

$$\begin{aligned} \psi\left(\frac{a}{s}\right) \cdot \psi\left(\frac{s}{1}\right) &= \psi\left(\frac{a}{1}\right) \\ \psi\left(\frac{a}{s}\right) &= \psi\left(\frac{a}{1}\right) \cdot \psi\left(\frac{s}{1}\right)^{-1} \end{aligned}$$

And we can see $\psi\left(\frac{a}{s}\right)$ is uniquely defined! □

Exercise 2.E

We follow the hint from author: Define

$$S^{-1}M = \left\{ \frac{m}{s} \mid m \in M, s \in S \right\}.$$

And $S^{-1}M$ meet the requirement to become a module by: $\forall m_1, m_2 \in M, s_1, s_2 \in S$:

$$\frac{m_1}{s_1} = \frac{m_2}{s_2} \iff \exists s \in S, s(s_2m_1 - s_1m_2) = 0.$$

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{m_1s_2 + m_2s_1}{s_1s_2}.$$

And the $S^{-1}A$ module structure

$$\forall a \in A, s_1, s_2 \in S, m \in M, \frac{a}{s_1} \cdot \frac{m}{s_2} = \frac{am}{s_1s_2}.$$

Also, we define $M \Rightarrow S^{-1}M$ by

$$m \longrightarrow \frac{m}{1}.$$

Now for any A -module maps $M \rightarrow N$, we'll prove ϕ satisfies the universal property, i.e. the map $S^{-1}M \rightarrow N$, we denote it by φ , is unique: $\forall \frac{m}{s} \in S^{-1}M$, notice:

$$\begin{cases} \alpha(m) \in N \\ \alpha(s)^{-1} \in N \end{cases}.$$

Therefore we have $\varphi(\frac{m}{s}) = \alpha(m)\alpha(s)^{-1}$. So the ϕ in the problem satisfy the uniqueness of universal property, so it exist. $\square$

Exercise 2.F

1. Intuitively, we can let

$$\frac{(m_1, m_2, \dots, m_n)}{s} \longrightarrow \left(\frac{m_1}{s}, \frac{m_2}{s}, \dots, \frac{m_n}{s} \right).$$

On the other hand, inductively we only have to consider the case $n = 2$, and we let

$$\left(\frac{m_1}{s_1}, \frac{m_2}{s_2} \right) \longrightarrow \left(\frac{m_1s_2}{s_1s_2}, \frac{m_2s_1}{s_1s_2} \right).$$

And we're done after induction.

2. Notice that in the case of direct sum, all but finitely many coordinates is zero. Therefore the proof is exactly the same as the case in (a).
3. Counterexample: consider group of modules $\{M_n\}_{n=1}^\infty$, and set of prime numbers $P = \{p_n\}_{n=1}^\infty$. We consider fraction of $\{M_n\}_{n=1}^\infty$ by positive integers, we cannot map $\frac{m_1}{p_1}, \frac{m_2}{p_2}, \dots$ to a proper element in $S^{-1}(\bigotimes_{k \in I} M_k)$ since $p_1, p_2, \dots$ cannot have a shared denominator.

Exercise 2.G

More generally, we'll prove that

$$\forall m, n \in \mathbb{Z}^+, \mathbb{Z}/(m) \otimes_{\mathbb{Z}} \mathbb{Z}/(n) \cong \mathbb{Z}/(\text{Gcd}(m, n)).$$

First, we know $\mathbb{Z}/(m) \otimes_{\mathbb{Z}} \mathbb{Z}/(n)$ is generated by $1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)}$. Which means that every elements of $\mathbb{Z}/(m) \otimes_{\mathbb{Z}} \mathbb{Z}/(n)$ can be written in the form of $k \cdot (1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)})$, $k \in \mathbb{Z}^+$. Also, $\forall i, j \in \mathbb{Z}^+$,

$$k \cdot (1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)}) = (mi + nj) \cdot k \cdot (1_{\mathbb{Z}/(m)} \otimes 1_{\mathbb{Z}/(n)}).$$

And the proof can be done by Euclidean algorithm. $\square$

Exercise 2.H [Proved in Atiyah]**Exercise 2.I [Trivial]**

Because from the textbook content right above this question, we see any two maps T, T' has a unique A -linear map to each other.

Exercise 2.J

Because the construction in §1.2.5 is a bilinear map, is unique up to isomorphism.

Exercise 2.K

1. • $\forall b_1, b_2 \in B, b \otimes_A m \in B \otimes M$, we have

$$\begin{aligned} (b_1 + b_2)(b \otimes_A m) &= (b_1 + b_2)b \otimes_A m \\ &= (b_1b + b_2b) \otimes_A m \\ &= b_1b \otimes_A m + b_2b \otimes_A m \\ &= b(b_1 \otimes_A m) + b(b_2 \otimes_A m). \end{aligned}$$

$$\begin{aligned} (b_1b_2)(b \otimes_A m) &= (b_1b_2b) \otimes_A m \\ &= b_1(b_2(b \otimes_A m)). \end{aligned}$$

$$b(b_1 \otimes_A m + b_2 \otimes_A m) = b(b_1 \otimes_A m) + b(b_2 \otimes_A m).$$

- Denote such map by $\mathcal{F}$. We see for ring A and $B \otimes_A M \in \text{Mod}_B$,

$$\mathcal{F} : A \mapsto B \otimes_A M.$$

And for $\varphi \in \text{Mor}(A)$, $\varphi : A \mapsto A'$,

$$\mathcal{F}(\varphi) : B \otimes_A M \mapsto B \otimes_{A'} M.$$

2. We define addition as formal sum and multiplication as

$$(b_1 \otimes c_1) \cdot (b_2 \otimes c_2) = (b_1b_2) \otimes (c_1c_2).$$

And we can easily exam the properties to let it becomes a ring.

Exercise 2.L

We can define

$$(S^{-1}A) \otimes_A M \longrightarrow S^{-1}M.$$

As

$$\left(\frac{a}{s}\right) \otimes m \mapsto \frac{am}{s}.$$

And we can easily check that it meets the requirement to become a isomorphism.

Exercise 2.M

Because I only has finite nonzero elements, so the properties of tensor product can be verified easily.

Exercise 2.N

It's because

$$\alpha \circ \text{pr}_X = \beta \circ \text{pr}_Y.$$

Exercise 2.O

Intersection.

Because morphisms are inclusion embeddings.

Exercise 2.P

Denote the natural maps

$$\begin{aligned}\mathrm{pr}_X &: X \times Y \longrightarrow X. \\ \mathrm{pr}_Y &: X \times Y \longrightarrow Y.\end{aligned}$$

And the unique map

$$\begin{aligned}\alpha &: X \longrightarrow Z. \\ \beta &: Y \longrightarrow Z.\end{aligned}$$

Now we have to prove

$$\alpha \circ \mathrm{pr}_X = \beta \circ \mathrm{pr}_Y.$$

Which do work since Z is a final diagram, from which we know, up to isomorphisms, has only a single morphism sending $X \times Y$ to Z .

Exercise 2.Q

For notations, let

$$\begin{array}{ccc} U & \xrightarrow{f_1} & V \\ \downarrow g_1 & & \downarrow h_1 \\ W & \xrightarrow{f_2} & X \\ \downarrow g_2 & & \downarrow h_2 \\ Y & \xrightarrow{f_3} & Z \end{array}.$$

And we can see

$$\begin{aligned}h_2 \circ h_1 \circ f_1 &= h_2 \circ f_2 \circ g_1 \\ &= f_3 \circ g_2 \circ g_1.\end{aligned}$$

□

Exercise 2.R

$$\begin{array}{ccccc} X_1 \times_Y X_2 & & & & \\ & \searrow \exists! & & \searrow & \\ & X_1 \times_Z X_2 & \longrightarrow & X_2 & \\ & \downarrow & & \downarrow & \\ & X_1 & \longrightarrow & Y & \\ & & & \searrow & \\ & & & Z & \end{array}.$$

Because from the problem we have the commutative diagram

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_1 & \dashrightarrow & Z \end{array}.$$

And by the definition of fibered product, there must exist a unique factoring map $X_1 \times_Y X_2 \longrightarrow X_1 \times_Z X_2$.

Exercise 2.S

($Y \longrightarrow Y \times_Z Y$ is defined by identical embedding)

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & Y \\ & & \searrow \\ & & Z \end{array}$$

We need to prove the diagram below is a Cartesian Square:

$$\begin{array}{ccc} X_1 \times_Y X_2 & \longrightarrow & X_1 \times_Z X_2 \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y \times_Z Y \end{array}$$

Which becomes trivial when we notice

$$\begin{array}{ccccc} X_1 \times_Y X_2 & & & & \\ \searrow \exists! & \searrow & & \searrow & \\ & X_1 \times_Z X_2 & \longrightarrow & X_2 & \\ & \downarrow & & \downarrow & \\ & X_1 & & Y \times_Z Y & \longrightarrow Y \\ & \searrow & & \downarrow & \parallel \\ & & & Y & \longrightarrow Z \end{array}$$

Exercise 2.T

For

$$\begin{array}{ccccc} & & S & & \\ \psi_1 \nearrow & & \uparrow & \nwarrow \psi_2 & \\ S_1 & \xrightarrow{\varphi_1} & S_1 \sqcup S_2 & \xleftarrow{\varphi_2} & S_2 \end{array}$$

For $s_1 \in S_1, s_2 \in S_2$, define

$$\varphi_1(s_1) = (s_1, \cdot), \varphi_2(s_2) = (\cdot, s_2).$$

Now for any ψ_1, ψ_2 , we naturally have the induced morphism

$$\Psi : S_1 \sqcup S_2 \longrightarrow S$$

$$\Psi : (s_1, \cdot) \longmapsto \psi_1(s_1)$$

$$\Psi : (\cdot, s_2) \longmapsto \psi_2(s_2).$$

Which is unique up to isomorphism while making the diagram commutes.

Exercise 2.U

Denote

$$\begin{array}{ccccc} D & & & & \\ \swarrow \varphi & \searrow h & & & \\ & B \otimes_A C & \longleftarrow & C & \\ \uparrow g & \uparrow & & \uparrow & \\ B & \longleftarrow & A & & \end{array}$$

and we obtain the unique map φ by

$$\varphi((a \cdot b) \otimes c) = g(b)$$

$$\varphi(b \otimes (a \cdot c)) = h(c)$$

Exercise 2.V

$$\begin{array}{ccccc} A & & & & \\ \downarrow \leq 1 & \downarrow \leq 1 & & \downarrow \leq 1 & \\ X & \xrightarrow{\alpha} & Y & \xrightarrow{\beta} & Z \end{array}$$

From the graph we obtain the proof easily.

Exercise 2.W

- When π is a monomorphism: For

$$\begin{array}{ccc} X \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow \pi \\ X & \xrightarrow{\pi} & Y \end{array}$$

From the definition of monomorphism, we know there's only one morphism in $\text{Mor}(X \times_Y X, X)$ making the diagram commutes. Therefore the fiber product must exist, where the morphism in between is an isomorphism.

- When $X \times_Y X$ exists, then any A letting the diagram below commute

$$\begin{array}{ccc} A & & \\ \swarrow \exists! & \searrow & \\ X \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow \pi \\ X & \xrightarrow{\pi} & Y \end{array}$$

whose morphism to $X \times_Y X$ will factor through $X \times_Y X$ uniquely. Therefore any two map:

$$\alpha, \beta : A \longrightarrow X$$

can be factorized into

$$\alpha = \alpha' \circ p, \beta = \beta' \circ p$$

Where $p : A \longrightarrow X \times_Y X$ and $\alpha', \beta' : X \times_Y X \longrightarrow X$.

$$\begin{array}{ccc} A & & \\ \swarrow p & \searrow \beta & \\ X \times_Y X & \longrightarrow & X \\ \downarrow \alpha' & & \downarrow \pi \\ X & \xrightarrow{\pi} & Y \end{array}$$

So if

$$\pi \circ \alpha = \pi \circ \beta$$

We'll have

$$\pi \circ \alpha' \circ p = \pi \circ \beta' \circ p$$

By the definition of diagonal morphism we know $\alpha' = \beta'$, therefore $\alpha' \circ p = \beta' \circ p$, i.e. $\alpha = \beta$. π satisfy the left cancellation law, so it's a monomorphism.

Exercise 2.X

$$\begin{array}{ccccc} & & X_1 \times_Y X_2 & & \\ & \searrow & & \searrow & \\ & & X_1 \times_Z X_2 & \longrightarrow & X_2 \\ & \searrow & \downarrow & & \downarrow \\ & & X_1 & \longrightarrow & Y \\ & & & & \searrow \\ & & & & Z \end{array}$$

From Exercise 1.2.R we know there exist a unique morphism

$$\varphi : X_1 \times_Y X_2 \longrightarrow X_1 \times_Z X_2.$$

Now since $Y \rightarrow Z$ is monomorphism, from the commutativity of $X_1 \times_Z X_2 \rightarrow X_1 \rightarrow Y \rightarrow Z$ and $X_1 \times_Z X_2 \rightarrow X_2 \rightarrow Y \rightarrow Z$ we can get the commutativity of

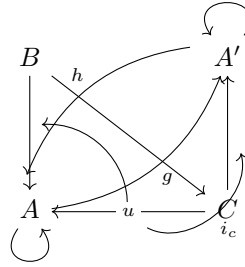
$$\begin{array}{ccc} X_1 \times_Z X_2 & \longrightarrow & X_2 \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & Y \end{array}$$

And therefore there's a unique morphism $\psi : X_1 \times_Z X_2 \rightarrow X_1 \times_Y X_2$. By their uniqueness, we have

$$\varphi \circ \psi \circ \varphi = \varphi, \quad \psi \circ \varphi \circ \psi = \psi$$

So $\varphi \circ \psi = \text{Id}_{X_1 \times_Z X_2}$, $\psi \circ \varphi = \text{Id}_{X_1 \times_Y X_2}$.

Exercise 2.Y



1. We simply let Id_C correspond to the morphism $A \rightarrow A'$ and verify the properties stated in the problems. Also, uniqueness can be obtained by letting $C = A$.
2. We follow from the hint: Let $C = A'$ and we get a bijection

$$\text{Id}_{A'} : (\text{Mor}(A', A)) \rightarrow (\text{Mor}(A', A'))$$

We pick out h from $\text{Mor}(A', A)$ such that $\text{Id}_{A'}(h) = \text{Id}_{A'}$. From **(a)** we know $\text{Id}_{A'}$ can be represented using g :

$$\text{Id}_{A'} = \text{Id}_{A'}(h) = g \circ h$$

On the other hand, we prove Id_A^{-1} can be induced by h using the similar method to **(a)**.

$$\exists! \text{Id}_A \in \text{Mor}(A, A), \quad \text{Id}_A(\text{Id}_A) = g \in \text{Mor}(A, A)$$

And we can see

$$\text{Id}_A = \text{Id}^{-1}(g) = h \circ g$$

□

Exercise 2.Z [Skipped temporarily due to the star.]
the morphisms $B \rightarrow A$.

1. We consider the contravariant functors of

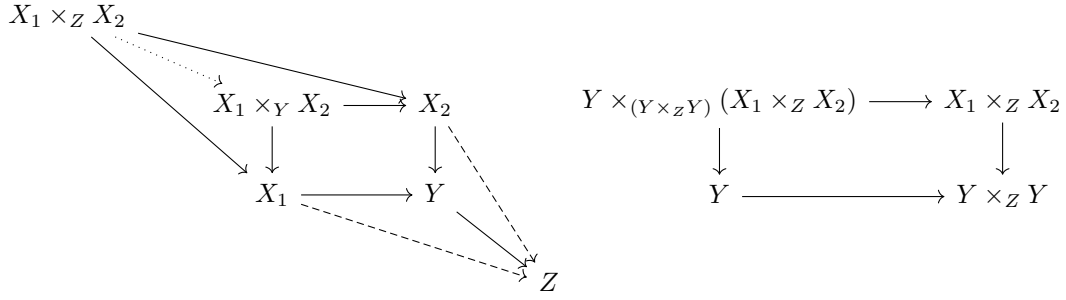
2.

3 Limits and colimits

Exercise 3.A [Trivial]

Since $\mathcal{I}$ is a partially ordered set, there's at most one morphism for any two object in $\mathcal{I}$.

Exercise 3.B



For $X_1 \times_Y X_2$:

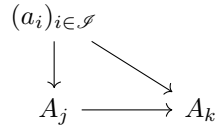
$$\begin{aligned} X_1 \times_Y X_2 \rightarrow Y &= X_1 \times_Y X_2 \rightarrow X_1 \rightarrow Y \\ &= X_1 \times_Y X_2 \rightarrow X_2 \rightarrow Y \\ X_1 \times_Y X_2 \rightarrow Z &= X_1 \times_Y X_2 \rightarrow Y \rightarrow Z \end{aligned}$$

For $Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2)$:

$$\begin{aligned} Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow Y &= Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow X_1 \times_Z X_2 \rightarrow X_1 \rightarrow Y \\ &= Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow X_1 \rightarrow Y \\ Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow Z &= Y \times_{(Y \times_Z Y)} (X_1 \times_Z X_2) \rightarrow Y \rightarrow Z \end{aligned}$$

□

Exercise 3.C



Because from the definition of $(a_i)_{i \in J}$, we have

$$F(m)(a_j) = a_k$$

So $(a_i)_{i \in J} A \rightarrow A_j \rightarrow A_k$ commutes with $(a_i)_{i \in J} \rightarrow A_k$.

□

Exercise 3.D

1. It means fractions converting into different denominators stands for the same rational number, which is obviously true.
2. Because we can view the set category as posets, where morphisms in between two objects are unique.

Exercise 3.E

The proof is done by definition.

Exercise 3.F

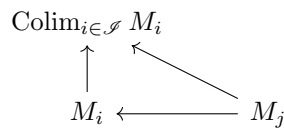
(Trivial, skipped.)

Exercise 3.G

(The proof is similar to 1.3.D I'll skip.)

Exercise 3.H

(Trivial, skipped.)



4 Adjoints

Exercise 4.A

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), B) & \xrightarrow{g^*} & \text{Mor}_{\mathcal{B}}(F(A), B') \\ \downarrow \tau_{AB} & & \downarrow \tau_{AB'} \\ \text{Mor}_{\mathcal{A}}(A, G(B)) & \xrightarrow{Gg^*} & \text{Mor}_{\mathcal{A}}(A, G(B')) \end{array}$$

Exercise 4.B

Use the natural property: For all $g : B' \rightarrow B$, we have

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), B') & \xrightarrow{g^*} & \text{Mor}_{\mathcal{B}}(F(A), B) \\ \downarrow \tau_{AB'} & & \downarrow \tau_{AB} \\ \text{Mor}_{\mathcal{A}}(A, G(B')) & \xrightarrow{Gg^*} & \text{Mor}_{\mathcal{A}}(A, G(B)) \end{array}$$

commutes. Therefore let $B' = F(A)$ and we'll have

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), F(A)) & \xrightarrow{g^*} & \text{Mor}_{\mathcal{B}}(F(A), B) \\ \downarrow \tau_{AF(A)} & & \downarrow \tau_{AB} \\ \text{Mor}_{\mathcal{A}}(A, GF(A)) & \xrightarrow{Gg^*} & \text{Mor}_{\mathcal{A}}(A, G(B)) \end{array}$$

commutes. Take $\text{Id}_{F(A)} \in \text{Mor}_{\mathcal{B}}(F(A), F(A))$ we would have

$$\begin{aligned} \tau_{AB}(g) &= \tau_{AB} \circ g \circ \text{Id}_{F(A)} \\ &= G(g) \circ \tau_{AF(A)} \circ \text{Id}_{F(A)}. \end{aligned}$$

Let $\eta_A = \tau_{AF(A)}(\text{Id}_{F(A)})$ and we're done.

On the other hand: vice versa. □

Exercise 4.C

For $\forall \beta \in \text{Bil}(M, N; P)$, we may define

$$\begin{aligned} f : M &\rightarrow \text{Hom}_A(N, P) \\ m &\mapsto \beta(m, (-)) \end{aligned}$$

i.e. $f(m)(n) = \beta(m, n)$.

For $\forall f \in \text{Hom}_A(M, \text{Hom}_A(N, P))$, let

$$\begin{aligned} \beta : M \times N &\rightarrow P \\ (m, n) &\mapsto f(m)(n) \end{aligned}$$

i.e. $\beta(m, n) = f(m)(n)$.

This defines the bijection. □

Exercise 4.D

Define F, G to be the functor $(\cdot) \otimes_A N, \text{Hom}_A(N, \cdot)$ respectively. We need to prove those τ_{XY} are natural, i.e. for $\forall \varphi : M \rightarrow M', \psi : P \rightarrow P'$

$$\begin{array}{ccc} \text{Hom}_A(M' \otimes_A N, P) & \xrightarrow{F\varphi^*} & \text{Hom}_A(M \otimes_A N, P) \\ \downarrow \tau_{M'P} & & \downarrow \tau_{MP} \\ \text{Hom}_A(M', \text{Hom}_A(N, P)) & \xrightarrow{\varphi^*} & \text{Hom}_A(M, \text{Hom}_A(N, P)) \end{array}$$

and

$$\begin{array}{ccc} \mathrm{Hom}_A(M \otimes_A N, P) & \xrightarrow{\psi^*} & \mathrm{Hom}_A(M \otimes_A N, P') \\ \downarrow \tau_{MP} & & \downarrow \tau_{MP'} \\ \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P)) & \xrightarrow{G\psi^*} & \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P')) \end{array}$$

commutes. This is obvious when we consider the property of homomorphisms. $\square$

Exercise 4.E

(I) τ_{NM} : For $\Phi : N \otimes_B A \rightarrow M$, define

$$\begin{aligned} \phi : N &\rightarrow M_B \\ n &\mapsto \Phi(n \otimes_B 1_A) \end{aligned}$$

τ_{NM}^{-1} : On the other hand, for $\phi : N \rightarrow M_B$, define

$$\Phi : N \otimes_B A \rightarrow M$$

by

$$\begin{array}{ccc} N \times A & \xrightarrow{\phi \times \mathrm{Id}} & M_B \times A \\ \downarrow \pi & & \searrow \\ N \times_B A & \xrightarrow{\Phi} & M \end{array}$$

and we have

$$\Phi : n \otimes_B a \mapsto a\phi(n)$$

Obviously the maps defined above are inverses to each other.

(II) We need to show such bijection (transformation) is natural, i.e. $\forall f \in \mathrm{Hom}_A(M, M')$,

$$\begin{array}{ccc} \mathrm{Hom}_A(N \otimes_B A, M) & \xrightarrow{f^*} & \mathrm{Hom}_A(N \otimes_B A, M') \\ \downarrow \tau_{NM} & & \downarrow \tau_{NM'} \\ \mathrm{Hom}_B(N, M_B) & \xrightarrow{(f^*)_B} & \mathrm{Hom}_B(N, M'_B) \end{array}$$

commute, while $\forall g \in \mathrm{Hom}_B(N, N')$,

$$\begin{array}{ccc} \mathrm{Hom}_A(N' \otimes_B A, M) & \xrightarrow{g^* \otimes_B \mathrm{Id}_A} & \mathrm{Hom}_A(N \otimes_B A, M) \\ \downarrow \tau_{N'M} & & \downarrow \tau_{NM} \\ \mathrm{Hom}_B(N', M_B) & \xrightarrow{g^*} & \mathrm{Hom}_B(N, M_B) \end{array}$$

commutes.

Pick a homomorphism from the start, then track it through the diagram directly and we're done. $\square$

Exercise 4.F [Trivial]

Exercise 4.G [Follow the hint, trivial]

(It's just like constructing a fraction field of a ring.)

Exercise 4.H

From **Exercise 1.4.E**, for ring homomorphism

$$A \longrightarrow S^{-1}A$$

we have

$$\mathrm{Hom}_{S^{-1}A}(S^{-1}N, M) \cong \mathrm{Hom}_A(N, M_A)$$

(I suppose the rest is trivial.)

5 An introduction to abelian categories

Exercise 5.A [Trivial]**Exercise 5.B**

$$\begin{aligned} \sum_{i=1}^n (-1)^i \dim A^i &= \sum_{i=1}^n (-1)^i (\dim \mathrm{Ker} f^i + \dim \mathrm{Im} f^i) \\ &= \sum_{i=1}^n (-1)^i (\dim \mathrm{Ker} f^i - \dim \mathrm{Im} f^{i-1}) \\ &= \sum_{i=1}^n (-1)^i h^i(A^\bullet) \end{aligned}$$

□

Exercise 5.C

This is obtained immediately from $\mathcal{C}$ being abelian category.

□

Exercise 5.D

(i) By (1.5.6.3) we know any chain complex

$$\dots \longrightarrow A^{i-1} \xrightarrow{f^{i-1}} A^i \xrightarrow{f^i} A^{i+1} \xrightarrow{f^{i+1}} \dots$$

can be decomposed into series of

$$0 \longrightarrow \mathrm{Ker} f^i \longrightarrow A^i \longrightarrow \mathrm{Im} f^i \longrightarrow 0$$

and

$$0 \longrightarrow \mathrm{Im} f^{i-1} \longrightarrow \mathrm{Ker} f^i \longrightarrow H^i(A^\bullet) \longrightarrow 0.$$

Therefore the any homomorphism between chain complexes naturally induces homomorphisms such as

$$\mathrm{Ker} f^i \longrightarrow \mathrm{Ker} g^i$$

and

$$\mathrm{Im} f^i \longrightarrow \mathrm{Im} g^i.$$

Hence

$$\mathrm{Ker} f^i / \mathrm{Im} f^{i-1} \longrightarrow \mathrm{Ker} g^i / \mathrm{Im} g^{i-1}$$

i.e.

$$H^i(A^\bullet) \longrightarrow H^i(B^\bullet).$$

Done.

(ii) (I think it's trivial.)

□

Exercise 5.E

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & C^{i-1} & \xrightarrow{d_C^{i-1}} & C^i & \xrightarrow{d_C^i} & C^{i+1} \xrightarrow{d_C^{i+1}} \cdots \\
& & \downarrow & & \downarrow & & \downarrow \\
\cdots & \longrightarrow & D^{i-1} & \xrightarrow{d_D^{i-1}} & D^i & \xrightarrow{d_D^i} & D^{i+1} \xrightarrow{d_D^{i+1}} \cdots
\end{array}$$

This is equivalent to prove $f - g$ is the zero mapping on homology, i.e. $dw - wd$ gives the zero mapping on homology, which is trivial. □

Exercise 5.F [Trivial]

(Just decompose those sequences)

Exercise 5.G

- (a) Trivial.
- (b) (i) See [Introduction to Commutative Algebra, **Proposition 2.18** – M. Atiyah], trivial.
(ii)
- (c) (i) See [Introduction to Commutative Algebra, **Proposition 2.9** – M. Atiyah], trivial.
(ii)
- (d) (i) See [Introduction to Commutative Algebra, **Proposition 2.9** – M. Atiyah], trivial.
(ii)

Exercise 5.H

First, by

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow A^M \longrightarrow 0$$

being exact, we have

$$S^{-1}A^{\oplus q} \longrightarrow S^{-1}A^{\oplus p} \longrightarrow S^{-1}A^M \longrightarrow 0$$

being exact, hence

$$0 \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}A^{\oplus q}, S^{-1}N) \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}A^{\oplus p}, S^{-1}N) \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$$

is exact.

On the other hand,

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow A^M \longrightarrow 0$$

being exact implies

$$0 \longrightarrow \text{Hom}_A(A^{\oplus q}, N) \longrightarrow \text{Hom}_A(A^{\oplus p}, N) \longrightarrow \text{Hom}_A(M, N)$$

is exact, therefore

$$0 \longrightarrow S^{-1}\text{Hom}_A(A^{\oplus q}, N) \longrightarrow S^{-1}\text{Hom}_A(A^{\oplus p}, N) \longrightarrow S^{-1}\text{Hom}_A(M, N)$$

is exact.

It's easy to define a natural homomorphism $S^{-1}\text{Hom}_A(M, N) \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$ explicitly. Also, obviously

$$\text{Hom}_{S^{-1}A}(S^{-1}A, S^{-1}N) \cong S^{-1}N \text{ by } \bar{f} \mapsto f(1)$$

and

$$S^{-1}\text{Hom}_A(M, N) \cong S^{-1}N \text{ by } \frac{f}{s} \mapsto \frac{f(1)}{s}.$$

Therefore we have the direct sum

$$\text{Hom}_{S^{-1}A}(S^{-1}A^{\oplus r}, S^{-1}N) \cong (S^{-1}N)^{\oplus r} \cong S^{-1}\text{Hom}_A(M^{\oplus r}, N).$$

□

Exercise 5.I [Beautiful!]

(Note that in homological algebra, it's a consequence that every left, right exact, or exact, functor being additive. Therefore it's well-defined to write $H^\bullet F$.)

(i) We follow the hint: consider the exact sequence

$$C^i \xrightarrow{d^i} C^{i+1} \longrightarrow \text{Coker } d^i \longrightarrow 0$$

and apply it with functor F , we have

$$F(C^i) \xrightarrow{F(d^i)} F(C^{i+1}) \longrightarrow F \text{Coker } d^i \longrightarrow 0$$

being exact. But at the same time we have the exact sequence

$$F(C^i) \xrightarrow{F(d^i)} F(C^{i+1}) \longrightarrow \text{Coker } Fd^i \longrightarrow 0.$$

Hence $F \text{Coker } d^i \cong \text{Coker } Fd^i$.

From (1.5.6.4), we know

$$0 \longrightarrow \text{Im } d^i \longrightarrow C^{i+1} \longrightarrow \text{Coker } d^i \longrightarrow 0$$

is exact. Apply it with F ,

$$F \text{Im } d^i \longrightarrow F C^{i+1} \longrightarrow F \text{Coker } d^i \longrightarrow 0$$

is exact.

$$\begin{array}{ccccccc} & & & & \text{Coker } Fd^i & & \\ & & & & \uparrow & \downarrow & \\ F \text{Im } d^i & \xrightarrow{\quad} & F C^{i+1} & \xrightarrow{\quad} & F \text{Coker } d^i & \longrightarrow & 0 \\ & \searrow \text{dotted} & \nearrow & & & & \\ & & \text{Im } Fd^i & & & & \end{array}$$

and we have the epimorphism

$$F \text{Im } d^i \longrightarrow \text{Im } Fd^i \longrightarrow 0$$

Again from (1.5.6.4), we know

$$0 \longrightarrow H^i(C^\bullet) \longrightarrow \text{Coker } d^{i-1} \longrightarrow \text{Im } d^i \longrightarrow 0$$

is exact. Apply it with F ,

$$\begin{array}{ccccccc} F H^i(C^\bullet) & \longrightarrow & F(\text{Coker } d^{i-1}) & \longrightarrow & F(\text{Im } d^i) & \longrightarrow & 0 \\ \downarrow \text{dotted} & & \downarrow \sim & & \downarrow & & \\ 0 & \longrightarrow & H^i F(C^\bullet) & \longrightarrow & \text{Coker } Fd^{i-1} & \longrightarrow & \text{Im } Fd^i \longrightarrow 0 \end{array}$$

we obtain the natural morphism by universal property.

(ii) Similar to (i), we use the two short exact sequences in (1.5.6.3) and obtain

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Im } Fd^{i-1} & \longrightarrow & \text{Ker } Fd^i & \longrightarrow & H^i(F C^\bullet) \longrightarrow 0 \\ & & \downarrow & & \downarrow \sim & & \downarrow \text{dotted} \\ 0 & \longrightarrow & F \text{Im } d^{i-1} & \longrightarrow & F \text{Ker } d^i & \longrightarrow & F H^i(C^\bullet) \end{array}.$$

Done

(iii) If F is exact, the epimorphism

$$F\text{Im } d^i \longrightarrow \text{Im } Fd^i \longrightarrow 0$$

obtained in (i) becomes isomorphism

$$F\text{Im } d^i \cong \text{Im } Fd^i,$$

therefore the induced natural morphism is an isomorphism. The same story goes to (ii). Thus the morphism in (i) and (ii), denoted by α, β , are both isomorphisms. Now we'll prove they're actually inverses to each other:

Consider

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ker } F(d^i) & \longrightarrow & H^i F(C^\bullet) & \hookrightarrow & \text{Coker } F(d^{i-1}) \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \beta & & \downarrow \sim \\ 0 & \longrightarrow & F(\text{Ker } d^i) & \longrightarrow & F(H^i(C^\bullet)) & \hookrightarrow & F(\text{Coker } d^{i-1}) \longrightarrow 0 \\ & & \uparrow \sim & & \downarrow \alpha & & \downarrow \sim \\ 0 & \longrightarrow & \text{Ker } F(d^i) & \longrightarrow & H^i F(C^\bullet) & \hookrightarrow & \text{Coker } F(d^{i-1}) \longrightarrow 0 \end{array}$$

We have

$$(\text{mono}) \circ (\alpha \circ \beta) \circ (\text{epi}) = (\text{mono}) \circ \text{id}_{H^i F(C^\bullet)} \circ (\text{epi})$$

By the cancellation law of monomorphism and epimorphism, we have

$$\alpha \circ \beta = \text{Id}$$

□

Exercise 5.J

Chase the diagrams and obtain those commute diagrams:

$$\begin{array}{ccccc} & & i & & \\ & & \downarrow & \searrow & \\ \text{Ker } h_i & \xrightarrow{\quad} & A_i & \xrightarrow{\quad} & B_i \\ \uparrow & \nearrow \exists! & \uparrow & & \uparrow \\ \text{Lim Ker } h_i & \xrightarrow{\quad} & \text{Lim } A_i & \xrightarrow{\exists!} & \text{Lim } B_i \\ \downarrow \exists! & \nearrow & \downarrow & \nearrow 0 & \\ \text{Ker } (\text{Lim } A_i \rightarrow \text{Lim } B_i) & & & & \end{array}$$

$$\begin{array}{ccccccc} & & C & & & & \\ & & \downarrow & \searrow 0 & \searrow & & \\ \text{Lim } A_i & \xrightarrow{\quad} & \text{Ker } h_i & \xrightarrow{\quad} & A_i & \xrightarrow{\quad} & B_i \\ \uparrow & \nearrow \exists! & \downarrow \exists! & & \downarrow & & \downarrow \\ & & \text{Ker } h_j & \xrightarrow{\quad} & A_j & \xrightarrow{\quad} & B_i \\ \uparrow & & & & & & \\ \text{Ker } (\text{Lim } A_i \rightarrow \text{Lim } B_i) & & & & & & \end{array}$$

□

Exercise 5.K